

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
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MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

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Score: _____ Signature of the Lab Tutor: _____ Date: _____

Basic Router Configuration

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives:

- Understand the basic CLI commands used for router configuration
- Assign IP addresses to router interfaces.
- Enable and verify router interfaces
- Establish and test connectivity using ping and traceroute • Troubleshoot common configuration errors in router setup.
- Configure the login information.
- Configure the login password.

Keywords: Router

Duration: 03 hours

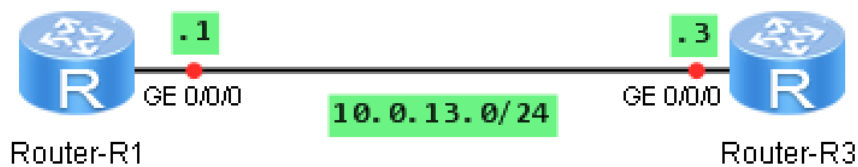
1 Overview of lab

1.1. Introduction

This lab focuses on the basic configuration of two routers to establish end-to-end communication within a network. Understanding of how to assign IP addresses to router interfaces, enable those interfaces, and implement static routing to ensure data packets can successfully travel between the networks connected to each router. This foundational exercise helps in understanding the core concepts of router setup, interface management, and routing mechanisms in a simulated network environment. The practical implementation enhances the understanding of real-world network communication and troubleshooting skills.

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

Topology



Scenario

A company has purchased two AR G3 routers that require commissioning before they can be used in the enterprise network. Items to be commissioned include setting device names, the system time, and password management.

Tasks

Step 1 View the system information

Run the display version command to view the software version and hardware information for the system.

```
<Huawei> display version
Huawei Versatile Routing Platform Software
VRP (R) software, Version 5.160 (AR2200 V200R007C00SPC600)
Copyright (C) 2011-2013 HUAWEI TECH CO., LTD
Huawei AR2220E Router uptime is 0 week, 3 days, 21 hours, 43 minutes
BKP 0 version information:
.....output omitted.....
```

The command output includes the VRP operating system version, device model, and startup time.

Step 2 Change the system time parameter

The system automatically saves the time. If the time is incorrect, run the **clock timezone** and **clock datetime** commands in the user view to change the system time.

```
<Huawei> clock timezone Local add 08:00:00
```

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

```
<Huawei> clock datetime 12:00:00 2016-03-11
```

The keyword Local can be exchanged with the current regional timezone name, and **add** replaced with **minus** where the timezone is west of UTC+0.

Run the **display clock** command to check that the new system time has taken effect.

```
<Huawei> display clock
2016-03-11 12:00:10
Friday
Time Zone(Local) : UTC+08:00
```

Step 3 Help features & Auto-completion functions

The question mark (?) is a wildcard, and the Tab is used as a shortcut to enter commands.

```
<Huawei> display ?
Cellular                               Cellular interface  aaa
AAA  access-user                       User access
accounting-scheme                     Accounting scheme   acl
<Group> acl command group  actual
Current actual  adp-ipv4                Ipv4
information  adp-mppls                  Adp-mppls
module  alarm                           Alarm
antenna                               Current antenna that outputting radio
anti-attack                               Specify anti-attack configurations  ap
<Group> ap command group  ap-auth-mode          Display AP
authentication mode  .....output omit.....
```

To display all the commands that start with a specific letter or string of letters, enter the desired letters and the question mark (?). The system displays all the commands that start with the letters entered. For example, if the string **dis?** is entered, the system displays all the commands that start with **dis**.

If a space exists between the character string and the question mark (?), the system will identify the commands corresponding to the string and display the parameters of the command. For example, if the string **dis ?** is entered and only the **display** command matches the **dis** string, the system displays the parameters of the display command. If multiple commands start with **dis**, the system displays an error.

The **Tab** key can also be pressed to complete a command. For example, if **dis** is entered followed by **Tab**, the system completes the **display** command. If multiple commands start with **dis**, the appropriate command can be selected.

If there are no other commands starting with the same letters, **dis** or **disp** can be entered to indicate **display**, and **int** or **inter** to indicate **interface**.

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

Step 4 Access the system view

Run the system-view command to access the system view to configure interfaces and protocols.

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]
```

Step 5 Change device names

To more easily identify devices, set device names during the device configuration. Change device names based on the lab topology, as shown below:

Change the name of the R1 router to **R1**.

```
[Huawei] sysname R1
[R1]
```

Change the name of the R3 router to **R3**.

```
[Huawei] sysname R3
[R3]
```

Step 6 Configure the login information

Configure the login information to indicate the login result.

```
[R1] header shell information "Welcome to the Huawei certification lab."
```

Run the preceding command to configure the login information. To check whether the login information has been changed, exit from the router command line interface, and log back in to view the login information.

```
[R1] quit
<R1> quit
```

```
Configuration console exit, please press any key to log on  Welcome
to the Huawei certification lab.
<R1>
```

Step 7 Configure console port parameters

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(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

The console port by default does not have a login password. Users must configure a password for the console port before logging in to the device.

The password can be changed in the password authentication mode to **huawei** in plain text. If there is no activity on the console port for the period of time specified by the timeout interval, the system will automatically log out the user. When this occurs, log in to the system again using the configured password.

The default timeout interval is set to 10 minutes. If a 10 minutes idle period is not a reasonable amount of time for the timeout interval, change the timeout interval to a more suitable duration, here this is set to 20 minutes.

```
[R1] user-interface console 0
[R1-ui-console0] authentication-mode password
[R1-ui-console0] set authentication password cipher
```

Warning: The "password" authentication mode is not secure, and it is strongly recommended to use "aaa" authentication mode.

Enter Password(<8-128>):

Confirm password:

```
[R1-ui-console0] idle-timeout 20 0
```

Run the **display this** command to check the configuration results.

```
[R1-ui-console0] display this
[V200R007C00SPC600]
#
user-interface con 0 authentication-mode
password
set authentication password cipher %$%$fIn'6>NZ6*~as(#J:WU%,#72Uy8cVlN^NXkT51E
^RX;>#75,%$%$ idle-timeout
20 0
```

Log out of the system and log back in, using the password set. It should be noted that this password is required to be set when the router is first initialized.

```
[R1-ui-console0] return
<R1> quit
```

```
Configuration console exit, please press any key to log on
Login authentication Password:
Welcome to Huawei certification lab
<R1>
```

Step 8 Configure interface IP addresses and descriptions

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Configure an IP address for the GigabitEthernet 0/0/0 interface of R1. The subnet mask can be configured using a dotted decimal format (255.255.255.0), or based on the subnet mask prefix length.

```
[R1] interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0] ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/0] description This interface connects to R3-G0/0/0
```

Run the **display this** command to check the configuration results at the current interface view.

```
[R1-GigabitEthernet0/0/0] display this
[V200R007C00SPC600]
#
interface GigabitEthernet0/0/0
  description This interface connects to R3-G0/0/0
  ip address 10.0.13.1 255.255.255.0
# return
```

Run the **display interface** command to view the interface description.

```
[R1] display interface GigabitEthernet0/0/0
GigabitEthernet0/0/0 current state : UP
Line protocol current state : UP
Last line protocol up time : 2016-03-11 04:13:09
Description: This interface connects to R3-G0/0/0
Route Port, The Maximum Transmit Unit is 1500
Internet Address is 10.0.13.1/24
IP Sending Frames' Format is PKTFMT_ETHNT_2, Hardware address is 5489-9876-830b
Last physical up time      :      2016-03-10 03:24:01  Last
physical down time       :      2016-03-10 03:25:29
Current system time: 2016-03-11 04:15:30
Port Mode: FORCE COPPER
Speed : 100, Loopback: NONE
Duplex: FULL, Negotiation: ENABLE
Mdi   : AUTO, Clock   : -
Last 300 seconds input rate 2296 bits/sec, 1 packets/sec
Last 300 seconds output rate 88 bits/sec, 0 packets/sec
Input peak rate 7392 bits/sec, Record time: 2016-03-10 04:08:41
Output peak rate 1120 bits/sec, Record time: 2016-03-10 03:27:56
Input: 3192 packets, 895019 bytes
  Unicast:          0,          Multicast:          1592
  Broadcast:        1600,        Jumbo:              0
Discard:            0,          Total Error:        0
CRC:                0,          Giants:           0
Jabbers:            0,          Throttles:       0
```

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```
Runts:                                0,                               Symbols:                                0
Ignoreds:                            0,                               Frames:                                0
Output: 181 packets, 63244 bytes
  Unicast:                            0,                               Multicast:                            0
Broadcast:                           181,                             Jumbo:                                0
Discard:                              0,                               Total Error:                          0
Collisions:                           0,                               ExcessiveCollisions:                  0
Late Collisions:                      0,                               Deferreds:                            0
  Input bandwidth utilization threshold : 100.00%
  Output bandwidth utilization threshold: 100.00%
  Input bandwidth utilization   : 0.01%
Output bandwidth utilization :    0%
```

The command output shows that the physical status and protocol status of the interface are **UP**, and the corresponding physical layer and data link layer are functional.

Once the status has been verified, configure the IP address and description for the interface of R3.

```
[R3] interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0] ip address 10.0.13.3255.255.255.0
[R3-GigabitEthernet0/0/0] description This interface connects to R1-G0/0/0
```

After completing the configuration, run the **ping** command to test the connection between R1 and R3.

```
<R1> ping 10.0.13.3
  PING 10.0.13.3: 56 data bytes, press CTRL_C to break
    Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=35 ms
  Reply from 10.0.13.3: bytes=56 Sequence=2 ttl=255 time=32 ms      Reply
  from 10.0.13.3: bytes=56 Sequence=3 ttl=255 time=32 ms          Reply from
  10.0.13.3: bytes=56 Sequence=4 ttl=255 time=32 ms              Reply from
  10.0.13.3: bytes=56 Sequence=5 ttl=255 time=32 ms
  --- 10.0.13.3ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 32/32/35 ms
```

Step 9 View the file list stored on the current device

Run the **dir** command in the user view to display the list of files in the current directory.

```
<R1> dir
Directory of flash:/
  Idx Attr   Size(Byte)   Date           Time (LMT)      FileName
```

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```
0 -rw- 1,738,816 Mar 10 2016 11:50:24 web.zip
1 -rw- 68,288,896 Mar 10 2016 14:17:5 ar2220E-v200r007c00spc600.cc
2 -rw- 739 Mar 10 2016 16:01:17 vrpcfg.zip
1,927,476 KB total (1,856,548 KB free)
```

```
<R3> dir
```

```
Directory of flash:/
```

Idx	Attr	Size(Byte)	Date	Time (LMT)	FileName
0	-rw-	1,738,816	Mar 10 2016	11:50:58	web.zip
1	-rw-	68,288,896	Mar 10 2016	14:19:0	ar2220E-v200r007c00spc600.cc
2	-rw-	739	Mar 10 2016	16:03:04	vrpcfg.zip

```
1,927,476 KB total (1,855,076 KB free)
```

Step 10 Manage device configuration files

Attempt to display the saved-configuration file.

```
<R1> display saved-configuration
```

```
There is no correct configuration file in FLASH
```

Since no save-configuration file exists, save the current configuration file.

```
<R1> save
```

```
The current configuration will be written to the device.
```

```
Are you sure to continue? (y/n) [n]:y
```

```
It will take several minutes to save configuration file, please wait.....
```

```
Configuration file had been saved successfully
```

```
Note: The configuration file will take effect after being activated
```

Run the following command again to view the saved configuration information:

```
<R1> display saved-configuration
```

```
[V200R007C00SPC600]
```

```
# sysname
```

```
R1
```

```
header shell information "Welcome to Huawei certification lab"
```

```
#
```

```
board add 0/1 1SA
```

```
board add 0/2 1SA
```

```
.....output omit.....
```

Run the following command to view the current configuration information:

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

```
<R1> display current-configuration
[V200R007C00SPC600]
# sysname
R1
  header shell information "Welcome to Huawei certification lab"
#
  board add 0/1 1SA
board add 0/2 1SA    board
add 0/3 2FE
.....output omit.....
```

A router can store multiple configuration files. Run the following command to view the configuration file to currently be used after the next startup:

```
<R3> display startup MainBoard:
  Startup system software:                flash:/ar2220E-V200R007C00SPC600.cc
Next startup system software:             flash:/ar2220E-V200R007C00SPC600.cc
  Backup system software for next startup: null
Startup saved-configuration file:         null
Next startup saved-configuration file:    flash:/vrpcfg.zip
  Startup license file:                   null
Next startup license file:                null
Startup patch package:                    null    Next
startup patch package:                    null    Startup
voice-files:                             null    Next startup
voice-files:                             null
```

Delete configuration files from the flash memory.

```
<R1> reset saved-configuration
This will delete the configuration in the flash memory.
The device configurations will be erased to reconfigure.
Are you sure? (y/n) [n]:y
Clear the configuration in the device successfully.
```

```
<R3> reset saved-configuration
This will delete the configuration in the flash memory.
The device configurations will be erased to reconfigure.
Are you sure? (y/n) [n]:y
Clear the configuration in the device successfully.
```

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

Step 11 Device restart procedure

Use the **reboot** command to restart the router.

```
<R1> reboot
Info: The system is now comparing the configuration, please wait.
Warning: All the configuration will be saved to the next startup configuration.
Continue ? [y/n]:n
System will reboot! Continue ? [y/n]:y
Info: system is rebooting ,please wait...
```

```
<R3> reboot
Info: The system is now comparing the configuration, please wait.
Warning: All the configuration will be saved to the next startup configuration.
Continue ? [y/n]:n
System will reboot! Continue ? [y/n]:y
```

The system asks to save the current configuration. It is necessary to determine whether the current configuration should be saved based on the requirements for the lab. If unsure as to whether the current configuration should be saved, do not save.

Final Configuration

```
[R1] display current-configuration
[V200R007C00SPC600]
# sysname
R1
  header shell information "Welcome to Huawei certification lab"
#
interface GigabitEthernet0/0/0
  description This interface connects to R3-G0/0/0
  ip address 10.0.13.1 255.255.255.0
#
user-interface con 0 authentication-
mode password
  set authentication password cipher
  %$%$4D0K*E"t/I7[{HD~kgW,%dgkQQ!&|;XTDq9SFQJ.27M%dj,%$%$
idle-timeout 20 0
# return
```

```
[R3] display current-configuration
[V200R007C00SPC600]
# sysname
R3
```

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

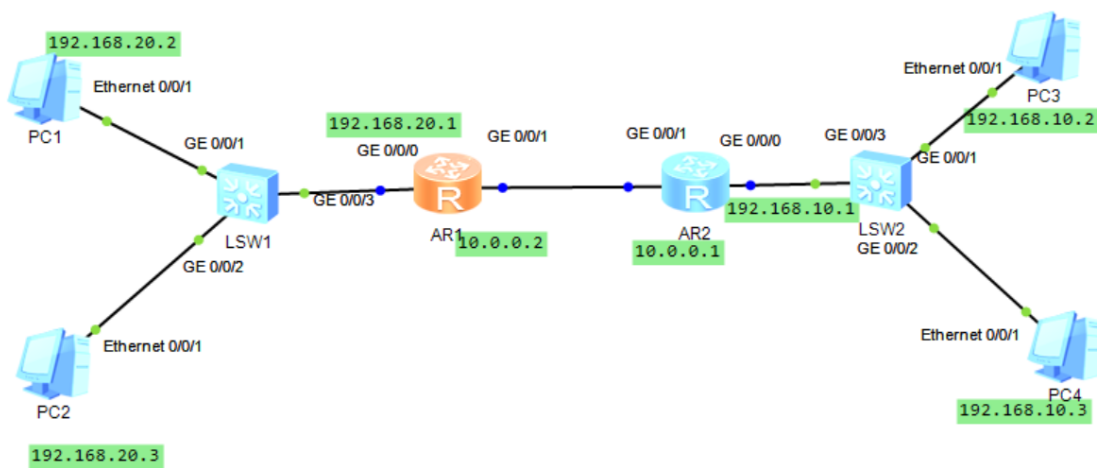
```
#
interface GigabitEthernet0/0/0
  description This interface connect to R1-G0/0/0
ip address 10.0.13.3 255.255.255.0
#
user-interface con 0
authentication-mode password set
authentication password cipher
%%$M8\H03:72:ERQ8JLoHU8,%t+lE:$9=a7"8%yMoARB]$B%t.,%%$ user-interface
vty 0 4
# return
```

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Network Protocol & Architecture
(6th SEMESTER, 4th Year) LAB EXPERIMENT # 02

Tasks

Try a different topology and demonstrate your learning.

Upload the `.topo` `.efz` `.cfg` files



LSW1

LSW2

AR2

AR1

```
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw)%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 192.168.20.1 255.255.255.0
#
interface GigabitEthernet0/0/1
ip address 10.0.0.2 255.255.255.0
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
ip route-static 192.168.10.0 255.255.255.0 10.0.0.1
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$!jneK]lkZ$Nx_K@hw`)0,"oEM]Qv)ir,o9ciPOWV
qSgJ"oH,%$%$
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[AR1]
```

LSW1	LSW2	AR2	AR1
------	------	-----	-----

```
#
set cpu-usage threshold 80 restore 75
#
aaa
authentication-scheme default
authorization-scheme default
accounting-scheme default
domain default
domain default_admin
local-user admin password cipher %$%$K8m.Nt84DZ)e#<0`8bmE3Uw)%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface GigabitEthernet0/0/0
ip address 192.168.10.1 255.255.255.0
#
interface GigabitEthernet0/0/1
ip address 10.0.0.1 255.255.255.0
#
interface GigabitEthernet0/0/2
#
interface NULL0
#
ip route-static 192.168.20.0 255.255.255.0 10.0.0.2
#
user-interface con 0
authentication-mode password
set authentication password cipher %$%$8h8K!PQOIK=n8n(@)XoK,"(Y$UESJ;n^H-6#UP'O
U5+9"(\,%$%$
user-interface vty 0 4
user-interface vty 16 20
#
wlan ac
#
return
[AR2]
```